

Hunters Hill High School
Mathematics Extension 2
Trial Examination, 2016



Hunters Hill
High School

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided
- The marks for each question are shown on the paper
- Show all necessary working in questions 11-16.

Total Marks: 100

Section I Pages 3-6
10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II Pages 7-14
90 marks

- Attempt Questions 11-16
- Allow about 2 hour 45 minutes for this section

Section I**10 marks Attempt Questions 1–10****Allow about 15 minutes for this section**

Use the multiple-choice answer sheet for Questions 1–10.

-
1. ω is a non-real root of the equation $z^5 + 1 = 0$.

Which of the following is not a root of this equation?

- (A) $\bar{\omega}$
(B) ω^2
(C) $\frac{1}{\omega}$
(D) ω^3

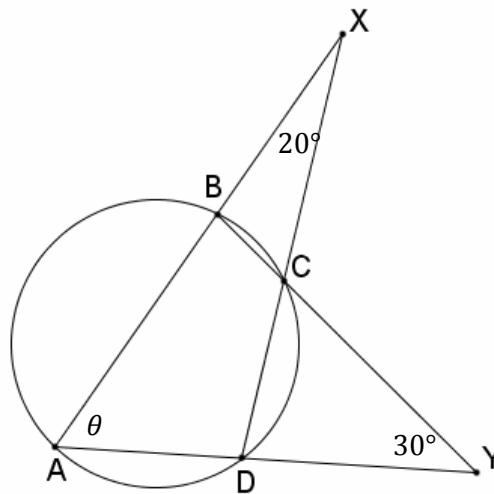
2. What is the acute angle between the asymptotes of the hyperbola $\frac{x^2}{3} - y^2 = 1$?

- (A) $\frac{\pi}{6}$
(B) $\frac{\pi}{4}$
(C) $\frac{\pi}{3}$
(D) $\frac{\pi}{2}$

3. What is the number of asymptotes on the graph of $f(x) = \frac{x^2}{x^2 - 1}$?

- (A) 1
(B) 2
(C) 3
(D) 4

4. The size of the angle θ in the diagram below is:



- (A) 50°
 (B) 55°
 (C) 60°
 (D) 65°
5. The locus of the graph of $\arg\left(\frac{z-2}{z+2i}\right) = \frac{\pi}{2}$ is
- (A) a semicircle passing through the origin
 (B) a circle with centre at the origin
 (C) an ellipse with a focus at the origin
 (D) a hyperbola not passing through the origin
6. The equation of the tangent of the ellipse $x = 3 \cos \theta$, $y = 2 \sin \theta$ at the point where $\theta = \frac{\pi}{3}$ is:
- (A) $6\sqrt{3}x - 4y - 5\sqrt{3} = 0$
 (B) $2x - 3\sqrt{3}y - 12 = 0$
 (C) $2x + 3\sqrt{3}y - 12 = 0$
 (D) $6\sqrt{3}x + 4y - 5\sqrt{3} = 0$

7. The base of a solid is the circle $x^2 + y^2 = 4$. Every cross section of the solid taken perpendicular to the x -axis is a right-angled, isosceles triangle with its hypotenuse lying on the base of the solid.

Which of the following is an expression for the volume of the solid?

- (A) $\frac{1}{4} \int_{-2}^2 (4 - x^2) dx$
- (B) $\int_{-2}^2 (4 - x^2) dx$
- (C) $2 \int_{-2}^2 (4 - x^2) dx$
- (D) $4 \int_{-2}^2 (4 - x^2) dx$

8. $\int x \sin 2x \, dx =$

- (A) $-\frac{x}{2} \cos 2x + \frac{1}{4} \sin 2x + C$
- (B) $-\frac{x}{2} \cos 2x - \frac{1}{4} \sin 2x + C$
- (C) $\frac{x}{2} \cos 2x + \frac{1}{4} \sin 2x + C$
- (D) $-2x \cos 2x + \sin 2x + C$

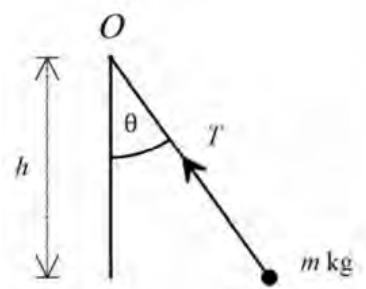
9. If $e^x + e^y = 1$, which of the following is an expression for $\frac{dy}{dx}$?

- (A) $-e^{x-y}$
- (B) e^{x-y}
- (C) e^{y-x}
- (D) $-e^{y-x}$

10. A particle of mass m moves in a horizontal circle with angular speed ω at a distance h below the point O .

Which of the following reflects the relationship between ω and h ?

- (A) $h = \omega^2 g$
- (B) $h = \omega g$
- (C) $g = \omega^2 h$
- (D) $g = \omega h$



End of Section I

Section II**90 marks****Attempt Questions 11–16****Allow about 2 hour and 45 minutes for this section**

Begin each question on a NEW SHEET of paper.

In questions 11 – 16 your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

- a. Factorise $z^2 + 2iz + 15$ 2
- b. Let $z = 1 + i$ and $u = 2 - i$. Find:
- i. $\operatorname{Im}(uz)$ 1
- ii. $|u - z|$ 1
- iii. $-i\bar{u}$ 1
- c. i. On an Argand diagram sketch the locus of z represented by $|z - 3| = 3$. 2
- ii. Explain why $\arg(z - 3) = 2 \arg z$ 1
- d. Find
- i. $\int \sec^3 x \tan x \, dx$ 2
- ii. $\int_4^7 \frac{dx}{x^2 - 8x + 19}$ 3
- e. If $z = \cos \theta + i \sin \theta$, use de Moivre's theorem to show that 2
- $$z^n + \frac{1}{z^n} = 2 \cos n\theta$$

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

a. Given the expression $\frac{6x^2 + 3x + 1}{(x + 1)(x^2 + 1)}$

i. Find numbers A, B and C such that

2

$$\frac{6x^2 + 3x + 1}{(x + 1)(x^2 + 1)} = \frac{A}{x + 1} + \frac{Bx + C}{x^2 + 1}$$

ii. Find $\int \frac{6x^2 + 3x + 1}{(x + 1)(x^2 + 1)} dx$

3

b. i. Express $-\sqrt{3} - i$ in modulus-argument form.

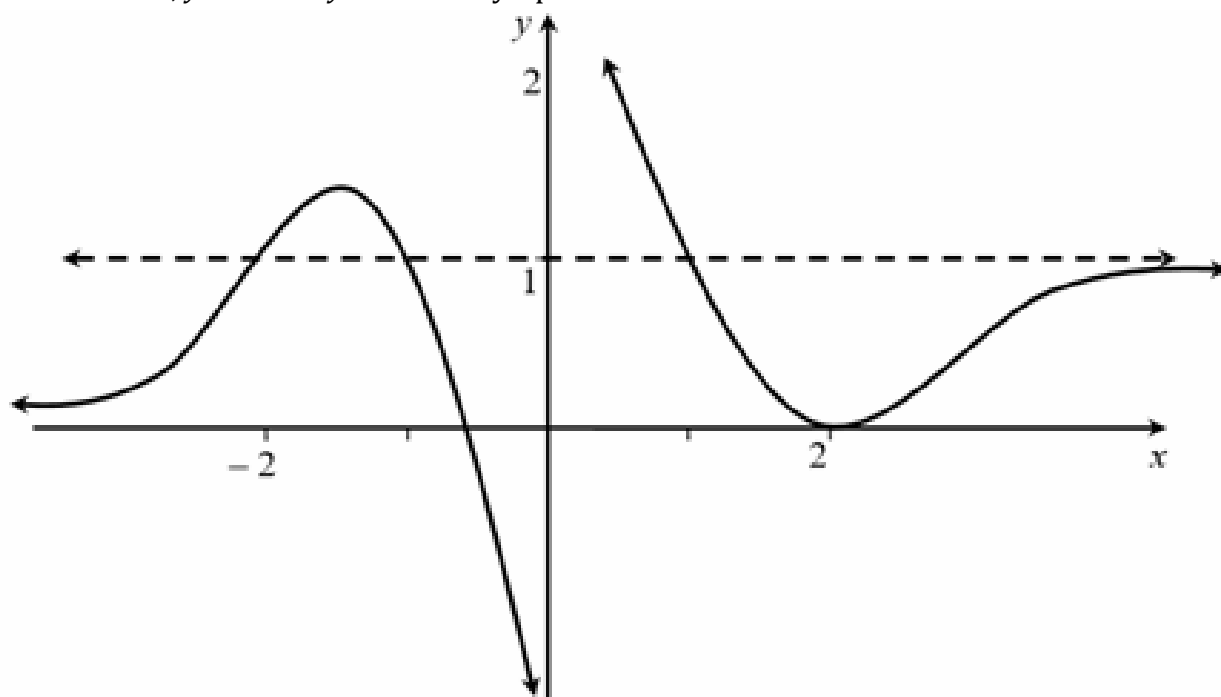
2

ii. Show that $(-\sqrt{3} - i)^6$ is a real number.

2

c. The diagram below is a sketch of the function $y = f(x)$.

The lines $x = 0, y = 0$ and $y = 1$ are asymptotes.



Draw separate one-third page sketches of the following curves, clearly indicating any important features such as turning points or asymptotes.

i. $y = \sqrt{f(x)}$

3

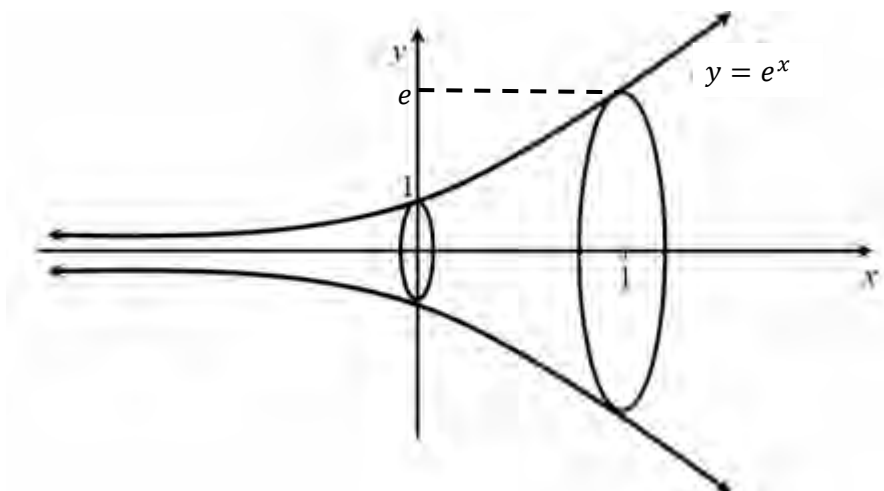
ii. $y = e^{f(x)}$

3

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- a. The arc defined by $y = e^x$, $0 \leq x \leq 1$, is rotated about the x -axis to form a curved bowl.



- i. Using the method of cylindrical shells, show that the volume, V , of the solid that makes the bowl is given by

3

$$V = \pi e^2 - 2\pi \int_1^e y \ln y \, dy$$

- ii. Find the volume, leaving your answer in exact form.

3

- iii. Hence, evaluate

1

$$\int_0^1 e^{2x} \, dx$$

- b. i. find all the 5th roots of -1 in modulus-argument form.

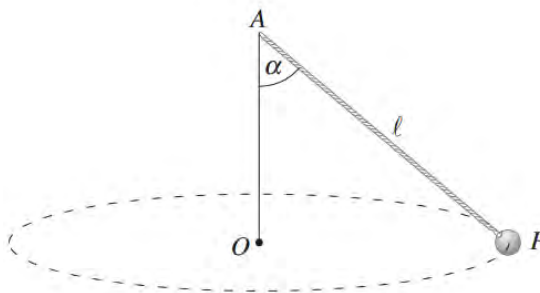
2

- ii. Sketch the 5th roots of -1 on an Argand diagram.

1

Question 13 continued on next page

- c. A particle P of mass m is attached by a string of length l to a point A . The particle moves with constant angular velocity ω in a horizontal circle with centre O which lies directly below A . The angle the string makes with OA is α .



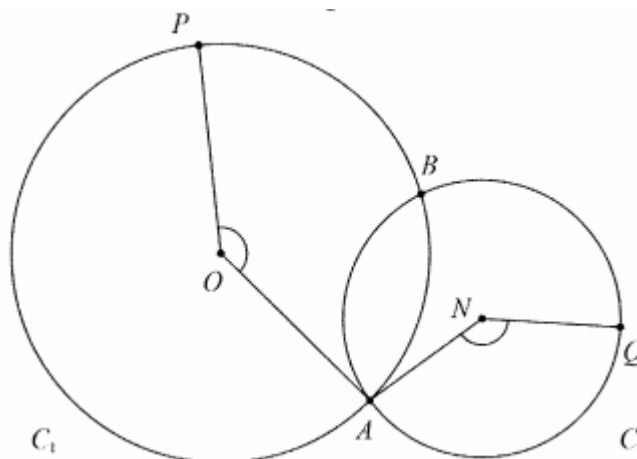
The forces acting on the particle are the tension, T , in the string and the force due to gravity, mg .

By resolving the forces acting on the particle in the horizontal and vertical directions, show that

$$\omega^2 = \frac{g}{l \cos \alpha}$$

3

- d. Two circles C_1 and C_2 with centres O and N respectively intersect at A and B . P lies on C_1 and Q lies on C_2 such that $\angle AOP = \angle ANQ$ and $\angle AOP > \angle AOB$, as shown in the diagram below.



3

Prove that the points P , B and Q are collinear.

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

a. i. Show that $\int_{-a}^0 f(x)dx = \int_0^a f(-x)dx$. 2

ii. Deduce that $\int_{-a}^a f(x)dx = \int_0^a \{f(x) + f(-x)\}dx$. 1

iii. Show that $\sec^2 x - \sec x \tan x = \frac{1}{1 + \sin x}$ 1

iv. Hence, or otherwise, evaluate $\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \frac{1}{1 + \sin x} dx$. 2

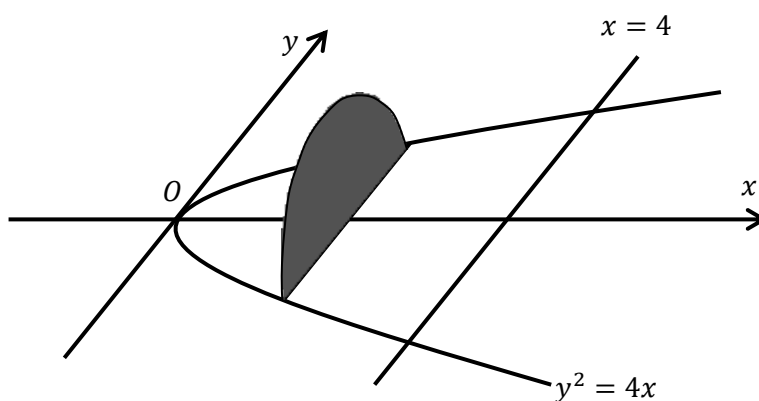
b. i. Prove that $a^2 + b^2 \geq 2ab$ 1

ii. Hence, or otherwise, prove that $(p + 2)(q + 2)(p + q) \geq 16pq$ where p and q are positive real numbers. 2

c. i. Express $(5 - i)^2(1 + i)$ in the form $a + ib$ where a and b are real. 1

ii. Hence, prove that $\tan^{-1} \frac{7}{17} + 2 \tan^{-1} \frac{1}{5} = \frac{\pi}{4}$. 2

d. The base of a solid is the region in the xy -plane enclosed by the parabola $y^2 = 4x$ and the line $x = 4$. Each cross section perpendicular to the x -axis is a semicircle.



Find the volume of the solid.

3

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- a. A body P of mass 0.5kg is suspended from a fixed point O by means of a light rod of length 1 metre .

The mass is rotated in a horizontal circle at a constant speed $v\text{ ms}^{-1}$.

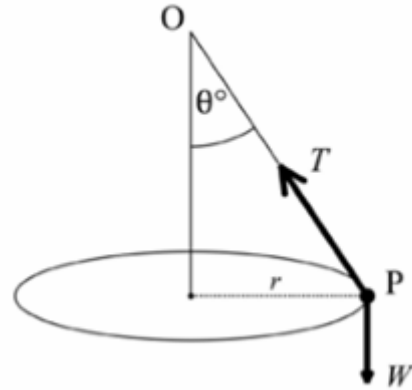
The rod makes an angle θ with the downward vertical direction as shown in the diagram below.

The tension in the rod is T newtons

and the weight of P is W newtons.

The radius of the circle is r metres.

Assume $g = 9.8\text{ ms}^{-2}$ and $\theta = 30^\circ$.



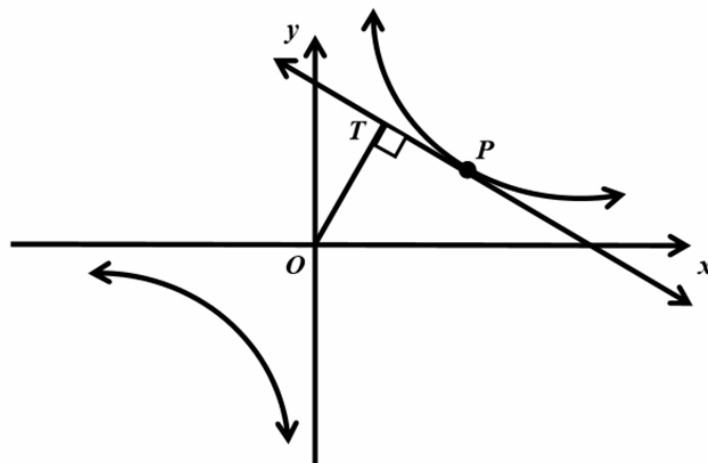
- | | | |
|------|--|---|
| i. | Show that $\tan \theta = \frac{v^2}{rg}$. | 3 |
| ii. | Find the tension T . | 1 |
| iii. | Find the speed $v\text{ ms}^{-1}$ of P . | 1 |
| iv. | Find the period of the motion. | 1 |

- b. Using the substitution $t = \tan \frac{x}{2}$, evaluate 4

$$\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \sin x}.$$

Question 15 continued on next page

c.



The point $P\left(cp, \frac{c}{p}\right)$ lies on the hyperbola $xy = c^2$. The point T lies at the foot of the perpendicular drawn from the origin O to the tangent at P .

- | | | |
|------|---|---|
| i. | Show that the tangent at P has equation $x + p^2y = 2cp$. | 2 |
| ii. | If the coordinates of T are (x_1, y_1) show that $y_1 = p^2x_1$. | 1 |
| iii. | Show that the locus of T is given by $(x^2 + y^2)^2 = 4c^2xy$. | 2 |

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

a. Consider the integral $I_n = \int_0^1 x^{2n+1} e^{-x^2} dx$.

i. Use integration by parts to show that $I_n = -\frac{1}{2e} + nI_{n-1}$, for $n \geq 1$. 2

ii. Show that $I_0 = \frac{1}{2} - \frac{1}{2e}$ and $I_1 = \frac{1}{2} - \frac{1}{e}$. 1

iii. Prove by mathematical induction that for all $n \geq 1$, 3

$$1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} = e - \frac{2eI_n}{n!}$$

iv. It is given that $0 \leq I_n \leq 1$ because $0 \leq x^{2n+1} e^{-x^2} \leq 1$, for $0 \leq x \leq 1$.
[Do NOT prove this]

Use this fact to help evaluate $\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots$, giving your answer in exact form. 1

b. An object on the surface of a liquid is released at time $t = 0$ and immediately sinks. Let x be its displacement in metres in a downward direction from the surface at time t seconds.

The equation of motion is given by

$$\frac{dv}{dt} = 10 - \frac{v^2}{40},$$

where v is the velocity of the object.

i. Show that $v = \frac{20(e^t - 1)}{e^t + 1}$. 4

ii. Use $\frac{dv}{dt} = v \frac{dv}{dx}$ to show that $x = 20 \log_e \left(\frac{400}{400 - v^2} \right)$. 2

iii. How far does the object sink in the first 4 seconds? 2

End of paper

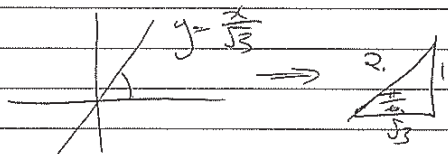
MXX-2016 TRIAL - SOLUTIONS

SECTION I

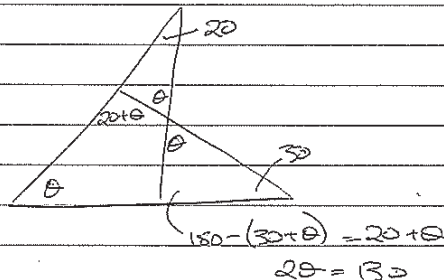
1. C

If ω is a 5th root, then
 $\omega^2, \omega^3, \omega^4, \omega^5$ are.
 also, roots of unity occur in
 conjugate pairs

2. C



3. C



4. D

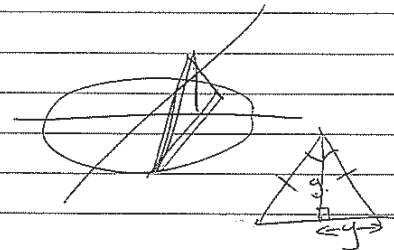
5. A

6. C

$$\begin{aligned} \frac{dx}{d\theta} &= -3\sin\theta, & \frac{dy}{d\theta} &= 2\cos\theta \\ \frac{dy}{dx} &= \frac{-2\cos\theta}{-3\sin\theta} & \left(= \frac{dy}{d\theta} \times \frac{d\theta}{dx} \right) \\ &= \frac{-2}{3} \cdot \frac{1}{\frac{\sin\theta}{\cos\theta}} \\ &= \frac{-2}{3\sqrt{3}} \end{aligned}$$

then perpendicular

7. B



$$\begin{aligned} A &= \frac{1}{2}(2y)y \\ &= y^2 \\ &= 4 - x^2 \end{aligned}$$

$$\delta V = \lim_{\delta x \rightarrow 0} \sum_{x=2}^4 (4-x^2) \delta x$$

8. A

$$\int x \sin 2x dx = x \left(-\frac{\cos 2x}{2} \right) - \int \left(-\frac{\cos 2x}{2} \right) dx$$

9. A

$$\begin{aligned} \frac{d}{dx} : e^x + e^y \frac{dy}{dx} &= 0 \\ \frac{dy}{dx} &= -\frac{e^x}{e^y} \\ &= -e^{x-y} \\ &= -e^{x-y} \end{aligned}$$

10. C

$$\begin{aligned} h &= \frac{g}{\omega^2} \\ \omega^2 h &= g \end{aligned}$$

11

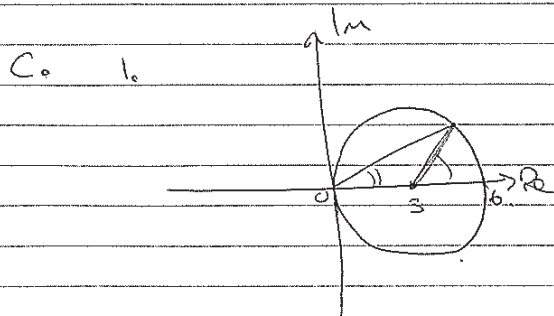
a. $z^2 + 2iz + 15 = z^2 + 2iz - 15i^2$ 2.
 $= (z + 5i)(z - 3i)$

b. $z = 1+i, u = 2-i$ 1.

i. $\ln(uz) = \ln((1+i)(2-i))$ 1.
 $= \ln(2+2i-i+1)$
 $= \ln(3+i)$
 $= \ln 3$

ii. $|u-z| = |2-i-(1+i)|$ 1.
 $= |2-i-1-i|$
 $= |1-2i|$
 $= \sqrt{1^2+2^2}$
 $= \sqrt{5}$

iii. $-i\bar{u} = -i(2+i)$ 1.
 $= -2i+1$
 $= 1-2i$



ii. Angle at centre is twice angle at circumference subtended by same arc

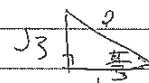
d. i. $I = \int \sec^2 x \cdot \sec x \cdot \tan x \, dx$

let $u = \sec x$
 $du = \sec x \cdot \tan x \, dx$

$I = \int u^2 du$
 $= \frac{1}{3} u^3 + c$

$= \frac{1}{3} \sec^3 x + c$

ii. $\int_4^7 \frac{dx}{x^2-8x+17} = \int_4^7 \frac{dx}{x^2-6x+16-3}$
 $= \int_4^7 \frac{dx}{(x-4)^2+3}$
 $= \int_4^7 \frac{1}{\sqrt{3}} \frac{dx}{1 + \frac{(x-4)^2}{3}}$
 $= \frac{1}{\sqrt{3}} \left(\tan^{-1} \frac{x-4}{\sqrt{3}} \right)_4^7$
 $= \frac{1}{\sqrt{3}} \left(\tan^{-1} \frac{7-4}{\sqrt{3}} - \tan^{-1} \frac{4-4}{\sqrt{3}} \right)$
 $= \frac{\pi}{3\sqrt{3}}$



e. $z^{\wedge} = \cos(n\theta) + i \sin(n\theta)$

$z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$

$= \cos(n\theta) - i \sin(n\theta)$

$z^{\wedge} + \frac{1}{z^{\wedge}} = \cos(n\theta) + i \sin(n\theta) + \cos(n\theta) - i \sin(n\theta)$
 $\frac{z^{\wedge}}{z^{\wedge}} = 2 \cos(n\theta)$

Q2.

a. i. $\frac{6x^2+3x+1}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$ 2.

$$6x^2+3x+1 = A(x^2+1) + (Bx+C)(x+1)$$

$$= x^2(A+B) + x(B+C) + (A+C)$$

comparing coefficients

$$A+B = 6 \quad -$$

$$B+C = 3 \quad -ii$$

$$A+C = 1 \quad -iii$$

i+iii

$$B-C = 5 \quad -iv$$

ii+iv

$$2B = 8$$

$$B = 4$$

$$\therefore C = -1$$

$$A = 2$$

$$\therefore A=2, B=4, C=-1$$

ii. $\int \frac{6x^2+3x+1}{(x+1)(x^2+1)} dx = \int \left(\frac{2}{x+1} + \frac{4x-1}{x^2+1} \right) dx$ 2.

$$= \int \left(\frac{2}{x+1} + \frac{4x}{x^2+1} - \frac{1}{x^2+1} \right) dx$$

$$= 2\ln(x+1) + 2\ln(x^2+1) - \tan^{-1}x + C$$

b i. $\sqrt{3}-i$

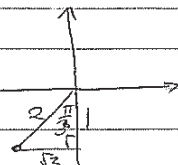
$$= 2 \operatorname{cis}\left(-\frac{\pi}{6}\right)$$

ii. $(-\sqrt{3}-i)^6 = 2^6 \operatorname{cis} 6\left(-\frac{5\pi}{6}\right)$

$$= 2^6 \operatorname{cis}(-5\pi)$$

$$= 2^6 [\cos(-5\pi) + i \sin(-5\pi)]$$

$$= -2^6$$

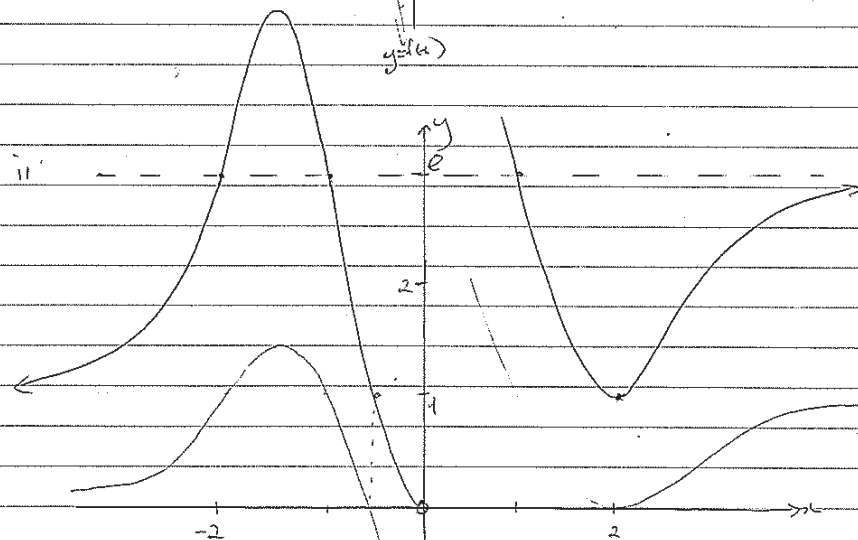
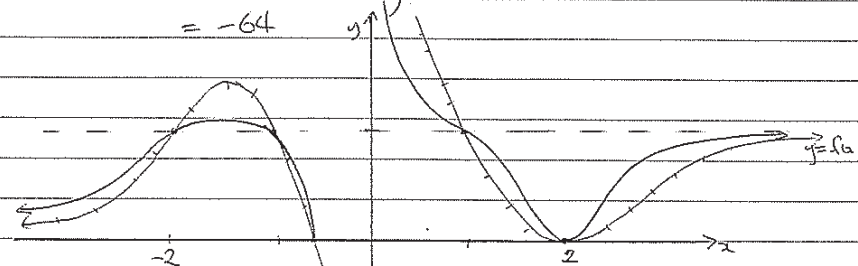


$$= 2^6 (\cos(-\pi) + i \sin(-\pi))$$

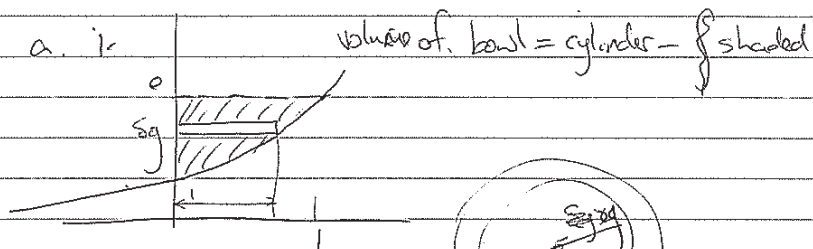
$$= 2^6 (-1 - 0i)$$

$$= -64$$

ci.



Q13. a. i.



Volume of bowl = cylinder - {shaded}

Shaded Volume

$$\begin{aligned} \delta A &= \pi((y+\delta y)^2 - y^2) \\ &= \pi(y^2 + 2y\delta y + \delta y^2 - y^2) \\ &= 2y\pi\delta y \end{aligned}$$

height(x) = $h y$

$$\delta V = 2\pi y \delta y \cdot h y$$

$$V = \lim_{\delta y \rightarrow 0} \sum_{y=1}^e 2\pi y h y \delta y$$

$$= 2\pi \int_1^e y \ln y dy$$

$$\text{cylinder} = \pi \cdot e^2 \cdot 1$$

$$\therefore \text{Volume of bowl, } V = \pi e^2 - 2\pi \int_1^e y \ln y dy$$

$$\begin{aligned} \text{ii } V &= 2\pi \int_1^e y \ln y dy = 2\pi \left[\frac{y^2}{2} \cdot \ln y \right]_1^e - \int_1^e \frac{y^2}{2} \cdot \frac{1}{y} dy \\ du &= y dy \\ dv &= \ln y dy \\ &= 2\pi \left(\frac{e^2 \cdot \ln e}{2} - \frac{1^2 \cdot \ln 1}{2} - \left[\frac{y^2}{4} \right]_1^e \right) \\ &= 2\pi \left(\frac{e^2}{2} - \frac{e^2}{4} + \frac{1}{4} \right) \\ &= \frac{\pi}{2} (e^2 + 1) \end{aligned}$$

$$\begin{aligned} V &= \pi e^2 - \frac{\pi}{2} (1 + e^2) \\ &= \frac{\pi e^2}{2} - \frac{\pi}{2} \\ &= \frac{\pi}{2} (e^2 - 1) \text{ units}^3 \end{aligned}$$

$$\text{iii) } \int_0^1 e^{2x} dx$$

Volume of bowl also equal to $\pi \int_0^1 (e^{2x})^2 dx$
when rotated about x-axis, slices perpendicular to axis

$$\therefore \pi \int_0^1 e^{2x} dx = \frac{\pi}{2} (e^2 - 1)$$

$$\therefore \int_0^1 e^{2x} dx = \frac{1}{2} (e^2 - 1)$$

$$= 3.194528049$$

b. i

$$\begin{aligned} z^5 &= -1 \\ z^5 &= r^5 \text{cis } 5\theta \\ -1 &= \text{cis } \pi \end{aligned}$$

equating

$$r^5 = 1$$

$$\therefore r = 1$$

$$\text{cis } 5\theta = \text{cis } \pi$$

$$5\theta = \pi + 2k\pi$$

$$\theta = \frac{\pi(2k+1)}{5}$$

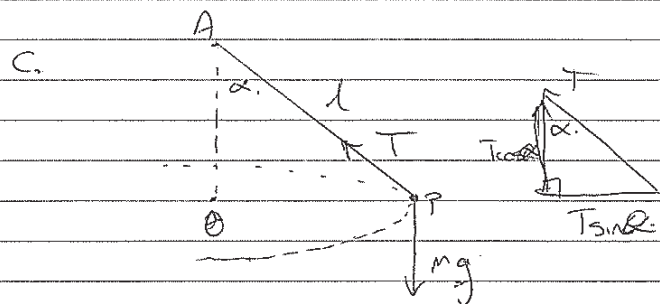
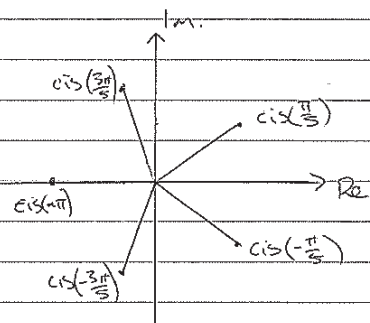
$$\text{for } k=0, z_1 = \text{cis } \frac{\pi}{5}$$

$$k=1, z_2 = \text{cis } \frac{3\pi}{5}$$

$$k=2, z_3 = \text{cis } \frac{5\pi}{5} = \text{cis } \pi$$

$$k=3, \quad z_4 = \cos \frac{3\pi}{5}$$

$$k=-1, \quad z_5 = \cos \left(-\frac{\pi}{5}\right)$$



Vertically: $T \cos \alpha = mg$ -i

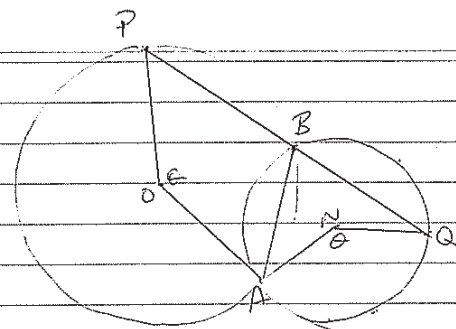
Radially: $T \sin \alpha = m r \omega^2$ -ii

$$\text{ii} \div \text{i} \quad \frac{T \sin \alpha}{T \cos \alpha} = \frac{m r \omega^2}{mg}$$

$$\frac{r}{l \cos \alpha} = \frac{r \omega^2}{g} \quad \sin \alpha = \frac{r}{l}$$

$$\therefore \omega^2 = \frac{g}{l \cos \alpha}$$

d.



$$\angle AOP = \angle ANQ \quad (\text{given})$$

$$= \theta$$

$$\angle BPA = (\text{obtuse } \angle AOP) \times \frac{1}{2} \quad (\text{angle at centre twice angle at circumference})$$

$$= (2\pi - \theta) \times \frac{1}{2}$$

$$= \pi - \frac{\theta}{2}$$

$$\angle QBA = \angle QNA \times \frac{1}{2} \quad (\text{angle at centre twice angle at circumference})$$

$$= \frac{\theta}{2}$$

$$\angle PBA + \angle QBA = \pi - \frac{\theta}{2} + \frac{\theta}{2}$$

$$= \pi$$

$\therefore P, B \text{ and } Q \text{ are collinear.}$

Q14.

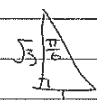
a. i. LHS = $\int_{-a}^0 f(x) dx$ 2.
 $= [F(x)]_{-a}^0$
 $= F(0) - F(-a)$

RHS = $\int_0^a f(x) dx$
 $= [-F(x)]_0^a$
 $= -F(a) - (-F(0))$
 $= F(0) - F(a)$
 $= \text{LHS}$

ii: $\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$ 1.
 $= \int_0^a f(x) dx - \int_0^a f(x) dx$
 $= \int_0^a \{f(x) - f(x)\} dx$

iii. LHS = $\sec^2 x - \sec x \tan x = \frac{1}{\cos^2 x}$ 1.
 $= \frac{1}{\cos^2 x} - \frac{1}{\cos x} \frac{\sin x}{\cos x}$
 $= \frac{1 - \sin^2 x}{\cos^2 x}$
 $= \frac{1 - \sin^2 x}{1 - \sin^2 x}$
 $= \frac{1 - \sin^2 x}{(1 + \sin x)(1 - \sin x)}$
 $= \frac{1}{1} = \text{RHS}$

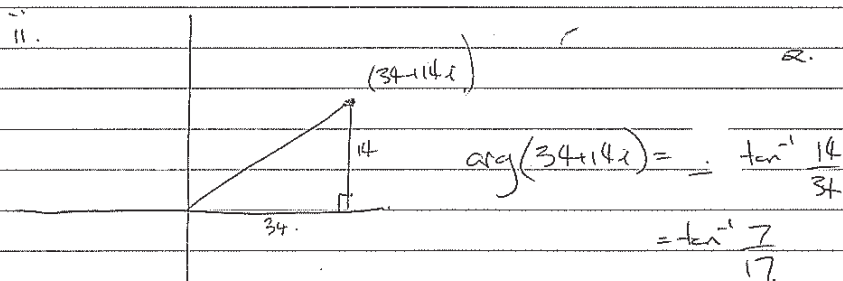
IV. $\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \frac{1}{1 + \sin x} dx = \int_0^{\frac{\pi}{6}} \left\{ \frac{1}{1 + \sin x} + \frac{1}{1 + \sin(-x)} \right\} dx$ 2.
 $= \int_0^{\frac{\pi}{6}} \{ \sec^2 x - \sec x \tan x + \sec^2(-x) - \sec(-x) \tan(-x) \} dx$
 $= \int_0^{\frac{\pi}{6}} \{ \sec^2 x - \sec x \tan x + \sec^2 x - \sec x(-\tan x) \} dx$
 $= \int_0^{\frac{\pi}{6}} 2 \sec^2 x dx$
 $= 2 \left[\tan x \right]_0^{\frac{\pi}{6}}$
 $= 2 \left(\tan \frac{\pi}{6} - \tan 0 \right)$
 $= 2 \left(\frac{1}{\sqrt{3}} - 0 \right)$
 $= \frac{2}{\sqrt{3}}$



b. i. $(a-b)^2 \geq 0$ 1.
 $a^2 - 2ab + b^2 \geq 0$
 $\therefore a^2 + b^2 \geq 2ab$

ii. $p+2 = (\sqrt{p})^2 + (\sqrt{2})^2 \geq 2\sqrt{p} \cdot \sqrt{2}$ 2.
 $q+2 = (\sqrt{q})^2 + (\sqrt{2})^2 \geq 2\sqrt{q} \cdot \sqrt{2}$
 $p+q \geq 2\sqrt{pq}$
 $\therefore (p+2)(q+2)(p+q) \geq 2\sqrt{p} \cdot \sqrt{2} \cdot 2\sqrt{q} \cdot \sqrt{2} \cdot 2\sqrt{pq}$
 $\geq 16\sqrt{pq}$

$$\begin{aligned}
 \text{c. i. } (5-i)^2(1+i) &= (25-10i+(-1))(1+i) \\
 &= (24-10i)(1+i) \\
 &= 24-10i+24i+10 \\
 &= 34+14i
 \end{aligned}$$



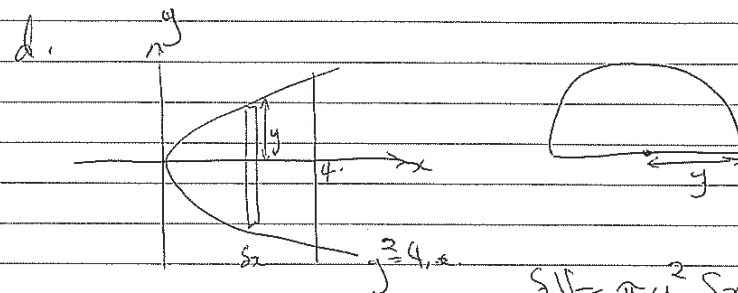
$$\arg(zw) = \arg(z) + \arg(w)$$

$$\therefore \arg(34+14i) = \arg(24-10i) + \arg(1+i)$$

$$= \arg(5-i)^2 + \arg(1+i)$$

$$\begin{aligned}
 \tan^{-1} \frac{7}{17} &= 2 \tan^{-1} \left(\frac{-1}{5} \right) + \tan^{-1} \frac{1}{1} \\
 &= -2 \tan^{-1} \frac{1}{5} + \frac{\pi}{4}
 \end{aligned}$$

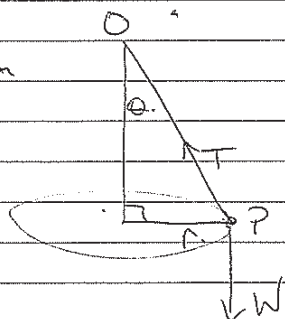
$$\therefore \tan^{-1} \frac{7}{17} + 2 \tan^{-1} \frac{1}{5} = \frac{\pi}{4}$$



$$\begin{aligned}
 \delta V &= \frac{\pi y^2}{2} \delta x \\
 &= \frac{\pi}{2} 4x \delta x \\
 &= 2\pi x \delta x
 \end{aligned}$$

$$\begin{aligned}
 V &= \lim_{\delta x \rightarrow 0} \sum_{x=0}^4 2\pi x \delta x \\
 &= 2\pi \int_0^4 x \, dx \\
 &= 2\pi \left[\frac{x^2}{2} \right]_0^4 \\
 &= 2\pi \left(\frac{4^2}{2} - \frac{0^2}{2} \right) \\
 &= 16\pi \text{ units}^3
 \end{aligned}$$

Q15a



$$m = 0.5 \text{ kg}$$

$$OP = 1 \text{ m}$$

$$W = mg$$

i.

Vertically: $T \cos \theta = mg$ (1)

radially: $T \sin \theta = \frac{mv^2}{r}$ (2)

$$(2) \div (1) \quad \frac{T \sin \theta}{T \cos \theta} = \frac{mv^2}{mg}$$

$$\therefore \tan \theta = \frac{v^2}{rg}$$

ii. from (1)

$$T \cos \theta = mg$$

$$T = \frac{mg}{\cos \theta}$$

$$= \frac{0.5 \times 9.8}{\cos 30^\circ}$$

$$= 5.658032638 \text{ Newtons}$$

iii.

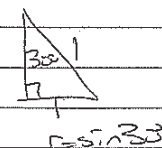
$$\tan \theta = \frac{v^2}{rg}$$

$$v^2 = rg \tan \theta$$

$$= \sin 30^\circ \times 9.8 \times \tan 30^\circ$$

$$= \frac{49}{105}$$

$$v = 1.68196799 \text{ ms}^{-1}$$



iv.

$$\omega = \frac{v}{r}$$

$$\text{period} = \frac{2\pi}{\omega}$$

$$= 2\pi r$$

$$= \frac{2\pi \sin 30^\circ}{1.68}$$

$$= 1.87807635 \text{ s.}$$

b.

$$* = \tan \frac{x}{2}$$

$$\sin x = \frac{2t}{1+t^2}$$

$$dx = \frac{1}{2} \sec^2 \frac{x}{2} dx$$

$$dx = 2dt \cdot \cos^2 \frac{x}{2}$$

$$= 2dt \cdot \frac{1}{1+t^2}$$

$$= \frac{2dt}{1+t^2}$$

also, when

$$x=0, t = \tan 0$$

$$= 0$$

$$x = \frac{\pi}{2}, t = \tan \frac{\pi}{4}$$

$$= 1$$

$$\begin{aligned}
 \int_0^{\frac{\pi}{2}} \frac{dx}{1+\sin x} &= \int_0^1 \frac{1}{1+\frac{2t}{1+t^2}} \cdot \frac{2dt}{1+t^2} \\
 &= \int_0^1 \frac{\cancel{1+t^2}}{1+t^2+2t} \cdot \frac{2dt}{\cancel{1+t^2}} \\
 &= \int_0^1 \frac{2dt}{(1+t)^2} \\
 &= 2 \left[\frac{(1+t)^{-1}}{-1} \right]_0^1 \\
 &= 2 \left(-\frac{1}{1+1} - \left(-\frac{1}{1+0} \right) \right) \\
 &= 2 \left(-\frac{1}{2} + 1 \right) \\
 &= 1
 \end{aligned}$$

C. i. $xy = c^2$

$$\frac{d}{dx}: \quad y + x \frac{dy}{dx} = 0$$

$$x \frac{dy}{dx} = -y$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$\begin{aligned}
 \text{at } P \quad m &= -\frac{y}{x} \\
 &= -\frac{1}{x}
 \end{aligned}$$

by point gradient
 $y - \frac{c}{p} = -\frac{1}{p^2} (x - cp)$

$$p^2 y - x = -x + cp$$

$$\therefore x + p^2 y = 2cp$$

ii. $OT \perp PT$
 so, gradient of OT is p^2 (m.m = -1)
 passing through (0,0)

$$\begin{aligned}
 \text{by point-gradient.} \\
 y - 0 &= p^2(x - 0) \\
 y &= p^2 x
 \end{aligned}$$

as T satisfies line
 $y_1 = p^2 x_1$

$$\begin{aligned}
 \text{iii} \quad x + p^2 y &= 2cp \\
 \text{also } p^2 &= \frac{y}{x}
 \end{aligned}$$

$$\text{so, } x + \frac{y}{x} \cdot y = 2c \sqrt{\frac{y}{x}}$$

$$x^2 + y^2 = 2c \sqrt{xy}$$

$$(x^2 + y^2)^2 = 4c^2 xy$$

16 a. a. $I_n = \int_0^1 x^{2n+1} e^{-x^2} dx$ 2.

i. $= \int_0^1 x^{2n} \cdot x e^{-x^2} dx$

let $u = x^{2n}$ $du = 2n x^{2n-1} dx$
 $dv = x e^{-x^2} dx$ $v = -\frac{1}{2} e^{-x^2}$

$$I_n = \left[x^{2n} \left(-\frac{1}{2} e^{-x^2} \right) \right]_0^1 - \frac{2n}{(-2)} \int_0^1 x^{2n-1} \cdot e^{-x^2} dx$$

$$= \left(-\frac{1^{2n}}{2} e^{-1} - 0 \right) + n \int_0^1 x^{2(n-1)+1} e^{-x^2} dx$$

$$= -\frac{1}{2e} + n I_{n-1}$$

ii. $I_0 = \int_0^1 -2x \cdot e^{-x^2} dx$

$$= -\frac{1}{2} \left[e^{-x^2} \right]_0^1$$

$$= -\frac{1}{2} (e^{-1} - e^{-0})$$

$$= -\frac{1}{2e} + \frac{1}{2}$$

$$= \frac{1}{2} - \frac{1}{2e}$$

∴ $I_1 = -\frac{1}{2e} + I_0$

$$= -\frac{1}{2e} + \frac{1}{2} - \frac{1}{2e}$$

$$= \frac{1}{2} - \frac{1}{e}$$

iii. Prove true for $n=1$

$$\text{LHS} = 1 + \frac{1}{1!} \quad \text{RHS} = e - \frac{2e \cdot I_1}{1!}$$

$$= 2$$

$$= e - 2e \left(\frac{1}{2} - \frac{1}{e} \right)$$

$$= e - \frac{1}{2}e + \frac{2e}{e}$$

$$= 2$$

$$= \text{LHS}$$

∴ true for $n=1$

Assume true for $n=k$

$$1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{k!} = e - \frac{2e I_k}{k!}$$

Prove true for $n=k+1$

$$\text{i.e. } 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{k!} + \frac{1}{(k+1)!} = e - \frac{2e I_{k+1}}{(k+1)!}$$

$$\text{LHS} = e - \frac{2e I_k}{k!} + \frac{1}{(k+1)!}$$

$$= e - \frac{2e}{k!} I_k + \frac{1}{(k+1)!} \cdot \frac{2e}{2e}$$

$$= e + \frac{2e}{(k+1)!} \left(-(k+1) I_k + \frac{1}{2e} \right) \quad \text{but } I_{k+1} = -\frac{1}{2e} + \frac{(k+1)!}{2e}$$

$$= e + \frac{2e}{(k+1)!} (-I_{k+1})$$

$$= e - \frac{2e I_{k+1}}{(k+1)!}$$

$$= \text{RHS}$$

∴ by induction, this is true.

$$10. \quad 0 \leq I_n \leq 1$$

$$\text{as } n \rightarrow \infty, \quad \frac{2e I_n}{n!} \rightarrow 0$$

$$\therefore 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots = e.$$

$$\text{and } \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots = e - 1$$

$$\text{b. i. } \frac{dv}{dt} = \frac{10 - v^2}{40}$$

$$= \frac{400 - v^2}{40}$$

$$\int \frac{40 dv}{400 - v^2} = \int dt$$

$$\frac{40}{(20-v)(20+v)} = \frac{A}{20-v} + \frac{B}{20+v}$$

$$40 = A(20+v) + B(20-v)$$

$$\text{at } v = 20,$$

$$40 = A(20+20) + B(0)$$

$$A = 1$$

$$\text{at } v = -20$$

$$40 = A(0) + B(20 - (-20))$$

$$B = 1$$

$$\int_0^{20} \left(\frac{1}{20-v} + \frac{1}{20+v} \right) dv = \dots = \int_0^t dt$$

$$\left[-\ln(20-v) + \ln(20+v) \right]_0^{20} = \left[t \right]_0^t$$

$$-\ln(20-20) + \ln(20+20) - (-\ln(20-0) + \ln(20+0)) = t - 0$$

$$\ln \frac{(20+v) \cdot 20}{(20-v) \cdot 20} = t$$

$$\frac{20+v}{20-v} = e^t$$

$$20+v = e^t(20-v)$$

$$= 20e^t - ve^t$$

$$ve^t + v = 20e^t - 20$$

$$v(e^t + 1) = 20(e^t - 1)$$

$$v = \frac{20(e^t - 1)}{e^t + 1}$$

$$\text{ii. } v \frac{dv}{dx} = \frac{10 - v^2}{40}$$

$$\int_0^{20} \frac{40 v dv}{400 - v^2} = \int_0^x dx$$

$$\left[-20 \ln(400 - v^2) \right]_0^{20} = x$$

$$x = -20 \ln(400 - 20^2) + 20 \ln(400 - 0)$$

$$= 20 \ln \left(\frac{400}{400 - 20^2} \right)$$

iii. at $t=4$

$$v = \frac{20(e^4 - 1)}{e^4 + 1}$$

$$= 19.2805516$$

$$x = 20 \ln \left(\frac{400}{400 - 19.28^2} \right)$$

$$= 5.300010989$$

$\therefore$ object sinks 5.3 m